

Quick Check

Lesson 1: The Blood Glucose Puzzle: Why Does Glucose Rise and Fall After Eating Ugali?

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 1.2: Chemicals of Life | Lesson 1

Wanjiru fasted, then ate ugali, sukuma wiki and beans. Her blood glucose was 4.5 mmol/L at 0 min and reached its highest value of 8.0 mmol/L at 60 min. By how much did her blood glucose rise from 0 to 60 min?

A 8.0 mmol/L

B 4.5 mmol/L

C 3.5 mmol/L

D 12.5 mmol/L

Wanjiru's blood glucose started at 4.5 mmol/L and rose by 3.5 mmol/L to a peak of 8.0 mmol/L. About what percentage above her starting level was the peak?

A About 56%

B About 3.5%

C About 78%

D About 44%

Wanjiru's blood glucose rose by 3.5 mmol/L between 0 and 60 min (4.5 to 8.0 mmol/L). It then fell by 3.5 mmol/L between 60 and 180 min (8.0 back to 4.5 mmol/L). What do these numbers show?

- A** The rise and fall cannot be compared because they have different start times.
- B** The rise was faster because the same 3.5 mmol/L change took 60 min going up but 120 min coming down.
- C** The fall was faster because glucose ended at a lower value than the peak.
- D** The rise and the fall were equally fast because both changes were 3.5 mmol/L.

A meal card for Wanjiru's lunch lists: ugali about 70 g carbohydrate, beans about 20 g protein and 15 g carbohydrate, sukuma wiki about 3 g protein plus fibre, vitamins and minerals, cooking oil about 10 g fat. Which food in the meal contained the largest amount of carbohydrate?

A Sukuma wiki

B Cooking oil

C Ugali

D Beans

After looking at the graph of blood glucose rising to 8.0 mmol/L and then falling back to 4.5 mmol/L, a group is asked what causes the fall after the peak. Which statement is the most honest scientific position at this stage of the unit?

- A** Glucose is burnt off in the stomach straight after the meal.
- B** The graph shows that glucose falls, but we have not yet explained what causes it, so it is a question to investigate.
- C** Glucose falls only because the person has stopped eating.
- D** Glucose simply disappears from the blood with time and nothing is involved.

Learners are writing questions for the Driving Question Board about the glucose graph. Which question is the most specific and can be investigated?

A Is glucose good?

B What chemicals are in ugali?

C Did Wanjiru enjoy her lunch?

D Why is food?

Learners draw an initial model of Wanjiru's meal and her glucose graph, using boxes and arrows with question marks where they are unsure. Which skeleton shows the correct order of the parts?

- A** Back to normal → ? → Food chemicals → ? → Glucose in blood
- B** Glucose in blood → ? → Food chemicals → ? → Back to normal
- C** Food chemicals → ? → Back to normal → ? → Glucose in blood
- D** Food chemicals → ? → Glucose in blood (rises to 8.0 mmol/L) → ? → Back to normal (4.5 mmol/L)

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).